

---

---

---

---

---



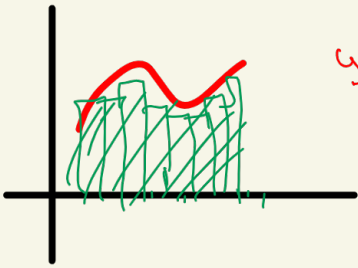
Jan 5, 2026

Part 1: Integrals

Part 2: Application of Integrals

Part 3: Sequences and Series

## Definite Integrals



$$y = f(x)$$

If  $f(x) > 0$

$$\int_a^b f(x) dx = \text{area below the graph}$$

Conceptual Interpretations =  
Continuous accumulation  
of quantity.

---

Infinite Integrals  $\int f(x)$

essentially antiderivatives

$x^2 \rightarrow$  antider.  $\frac{x^3}{3}$  so

the most general antider.

is  $\frac{x^3}{3} + C$ .

$\int f(x) dx = F(x) + C$  means

that most general antider.

of  $f(x)$

$$F'(x) = f(x) \quad \int_0^1 x^2 dx = \frac{1}{3}$$

$$\int x^2 dx = \frac{x^3}{3} + C$$

# FTC: Fundamental Theory of Calculus

$$\text{FTC: } \int_0^1 x^2 dx = \left[ \frac{x^3}{3} \right]_0^1 = \frac{1}{3} - \frac{0}{3} = \frac{1}{3}$$

this always works

January 7, 2026

$$(f(x)g(x))' = f'(x)g(x) + f(x)g'(x)$$

$$\int (f'(x)g(x) + f(x)g'(x)) dx = f(x)g(x) + C$$

$$\int f'(x)g(x) dx + \int f(x)g'(x) = f(x)g(x) + C$$

$$\int f(x)g'(x) = f(x)g(x) - \int f'(x)g(x) dx$$

Integration by parts

$$\int x^2 dx - 7 = \frac{x^3}{3} + (C-7) = \int x^2 dx$$

1)

$$\int x \cdot \cos x dx$$

Integration by  
parts

Solution:

$$\begin{aligned}
 \int x \cdot \cos x \, dx &= \int x \cdot (\sin x)' \, dx = \\
 &= x \cdot \sin x - \int x' \sin x \, dx = x \cdot \sin x - \\
 &- \int \sin x \, dx = x \cdot \sin x + \cos x + C
 \end{aligned}$$

$$\begin{aligned}
 2) \int 2^x \cdot x \, dx &= \int \left( \frac{2^x}{\ln 2} \right)' \cdot x \, dx = \\
 &= \frac{2^x}{\ln 2} \cdot x - \int \frac{2^x}{\ln 2} \cdot 1 \, dx =
 \end{aligned}$$

$$= \frac{2^x \cdot x}{\ln 2} - \frac{1}{\ln 2} \int 2^x \, dx \rightarrow \frac{2^x}{\ln 2}$$

$$= \frac{2^x \cdot x}{\ln 2} - \frac{2^x}{(\ln 2)^2}$$

$$\begin{aligned}
 3) \int x^2 e^x dx &= \int x^2 \cdot (e^x)' dx = \\
 &= x^2 e^x - \int (x^2)' e^x dx = x^2 e^x - 2 \int \\
 &= 2 \int \underbrace{x \cdot e^x}_{\text{blue box}} dx = x^2 e^x - 2 \int x \cdot (e^x)' dx \\
 &= x^2 \cdot e^x - 2 \left( x \cdot e^x - \int \boxed{x'}^1 e^x dx \right) = \\
 &= x^2 \cdot e^x - 2 \cdot x \cdot e^x + 2 \boxed{\int e^x dx} e^x \\
 &= \boxed{x^2 e^x - 2x e^x + 2 e^x + C}
 \end{aligned}$$

$$\begin{aligned}
 4) \int \ln x dx &= \int \boxed{1}^{x'} \cdot \ln x dx = \\
 &= \int x' \ln x dx = x \ln x - \int x (\ln x)' dx \\
 &= x \ln x - \int x \cdot \frac{1}{x} dx = x \ln x - \int 1 dx
 \end{aligned}$$

$$= \boxed{x \cdot \ln x - x + C}$$

$$5) \int \cos^2 x \, dx = \int \cos x \cdot (\sin x)' \, dx$$

$$= \cos x \cdot \sin x - \int (\cos x)' \sin x \, dx$$

$$= \cos x \cdot \sin x + \boxed{\int \sin^2 x \, dx}$$

$$= \cos x \cdot \sin x + \int (1 - \cos^2 x) \, dx$$

$$= \cos x \cdot \sin x + \int 1 \, dx - \int \cos^2 x \, dx$$

$$= \cos x \cdot \sin x + x - \int \cos^2 x \, dx$$

Solve for  $\int \cos^2 x \, dx$ :

$$2 \int \cos^2 x \, dx = \frac{1}{2} (\cos x \sin x + x) + C$$

$$u = x^2$$

$$u = g(x)$$

$$\frac{du}{dx} = 2x$$

$$g'(x)$$

$$du = 2x \, dx$$

$$g'(x) \, dx$$

$$\int \underbrace{f(x)}_u \underbrace{g'(x)}_{dv} \, dx = \underbrace{f(x)}_u \underbrace{g(x)}_v - \int \underbrace{g(x)}_v \underbrace{f'(x)}_{du}$$

$$\int u \, dv = uv - \int v \, du$$

1)

$$\int \underbrace{x}_u \cdot \underbrace{\cos x \, dx}_{dv} :$$

$$u = x \quad v = \sin x$$

$$dv = \cos x \cdot dx$$

$$du = dx$$



$$\int \underbrace{x}_u \cdot \underbrace{\cos x \, dx}_{dv} = \int u \, dv = uv - \int v \, du + C$$

$$= x \cdot \sin x - \int \sin x \, dx = x \cdot \sin x - \cos x + C$$


---

January 12, 2026

Substitution Rule (aka "u-sub")

$$\int (f(x) + g(x)) \, dx = \int f(x) \, dx + \int g(x) \, dx$$

$\int f(x) g(x) \, dx \rightarrow$  integration by parts

$$f(x) g(x) - \int f'(x) g(x) \, dx$$

\* often serves the purpose

$\int f(g(x)) \, dx \quad \text{X} \rightarrow$  no such thing

$\int f(g(x)) g'(x) \, dx \rightarrow$  there is a formula for this, substitution rule

$$\int f(g(x))g'(x) dx = \int \underbrace{F'(g(x))g'(x)}_{(F(g(x)))'} dx$$

$$= \int (F(g(x)))' dx = F(g(x)) + C \quad \text{Let } u = g(x)$$

$$= F(u) + C = \int f(u) du \quad \checkmark$$

$$\text{let } u = x^2$$

$$\frac{du}{dx} = 2x$$

$$du = 2x dx$$

$$u = g(x)$$

$$du = g'(x) dx$$

$$\frac{du}{dx} = g'(x)$$

$$\int \underbrace{f(g(x))}_u \underbrace{g'(x)}_{du} dx = \int f(u) du$$

# Practice

1.

$$\int x \cdot \cos(x^2) dx$$

$$\text{let } u = x^2$$

$$du = 2x dx \rightarrow \frac{du}{2}$$

$$\int x \cdot \cos(x^2) dx = \int \cos(u) \cdot \boxed{\frac{du}{2}} =$$

$$\frac{1}{2} \int \cos(u) du = \frac{1}{2} \sin(u) + C =$$

$$\frac{1}{2} \sin(x^2) + C$$

2.

$$\int x^2 \sqrt[3]{4x^3+3} dx$$

$$\text{let } u = 4x^3+3$$

$$du = 12x^2 dx \mid x^2 dx = \frac{du}{12}$$

$$\int x^2 \sqrt[3]{4x^3+3} = \int \sqrt[3]{u} \frac{du}{12} = \frac{1}{2} \int \sqrt[3]{u} du$$

$$= \frac{1}{12} \int u^{\frac{1}{3}} du = \frac{1}{12} \frac{u^{\frac{4}{3}}}{\frac{4}{3}} + C =$$

$$= \frac{u^{\frac{4}{3}}}{16} + C = \boxed{\frac{(4x+3)^{\frac{4}{3}}}{16}} + C$$

3.

$$\int \sin(10x) dx \quad \begin{array}{l} \text{let } u = 10x \\ du = 10 dx \\ dx = \frac{du}{10} \end{array}$$

$$\int \sin(10x) dx = \int \sin(u) \frac{du}{10} = \frac{1}{10} \int \sin(u) du$$

$$= -\frac{1}{10} \cos(u) + C = \boxed{-\frac{1}{10} \cos(10x) + C}$$

Useful shortcut:

If  $\int f(x) dx = F(x) + C$ , then

$$\cdot \int f(x+a) dx = F(x+a) + C$$

$$\int \cos(x+7) dx = \sin(x+7) + C$$

$$\int f(a \cdot x) dx = \frac{F(a \cdot x)}{a} + C$$

$$\int \cos(7 \cdot x) dx = \frac{\sin(7x)}{7} + C$$

$$5. \int \frac{x^3}{1+x^4} dx$$

Solution:

$$\text{Let } u = 1+x^4$$

$$du = 4x^3 dx \quad x^3 dx = \frac{du}{4}$$

$$\int \frac{x^3}{1+x^4} dx = \int \frac{du/4}{u} = \frac{1}{4} \int \frac{1}{u} du =$$

$$= \frac{1}{4} \ln|x| + C = \boxed{\frac{1}{4} \ln|1+x^4| + C}$$

$$6. \int \frac{x}{1+x^4} dx$$

$$\int \frac{x}{1+x^4} dx = \int \frac{x}{1+(x^2)^2} dx$$

$$\text{Let } u = x^2$$

$$du = 2x dx \quad x dx = \frac{du}{2}$$

$$\int \frac{1}{1+x^4} x dx = \int \frac{1}{1+u^2} \frac{du}{2} =$$

$$= \frac{1}{2} \int \frac{1}{1+u^2} du = \frac{1}{2} \tan^{-1}(u) + c$$

$$= \boxed{\frac{1}{2} \tan^{-1}(x^2) + c}$$

$$7. \int \frac{1}{x^2 + 14x + 130} dx$$

$$\text{Solution: } x^2 + 14x + 130 = (\quad)^2 + c$$

$$(a+b)^2 = a^2 + 2ab + b^2$$

$$x^2 + 14x + 130 = (x^2 + 2 \cdot x \cdot 7 + 7^2) + 7^2 + 130$$

$$= (x+7)^2 + 81 \rightarrow \text{Completing the square}$$

$$\int \frac{1}{(x+7)^2 + 81} dx \quad \text{Let } u = x+7$$

$$du = dx$$

$$\int \frac{1}{u^2 + 81} du = \int \frac{1}{u^2 + 9^2} du =$$

$$= \frac{1}{9} \tan^{-1}\left(\frac{u}{9}\right) + C = \boxed{\frac{1}{9} \tan^{-1}\left(\frac{x+7}{9}\right) + C}$$

11

$$\int \frac{\sin x \cdot \cos x}{1 + \sin^2 x} dx$$

$$\text{Let } u = \sin x \quad du = \cos x \, dx$$

$$\int \frac{\sin x \cos x}{1 + \sin^2 x} dx = \int \frac{u}{1+u^2} du$$

$$\text{Let } v = 1+u^2$$

$$dv = 2u du$$

$$u \cdot du = \frac{dv}{2}$$

$$\int \frac{1}{v} \frac{dv}{2} = \frac{1}{2} \int \frac{1}{v} dv = \frac{1}{2} \ln|v| + c$$

$$\frac{1}{2} \ln(1 + \sin^2 x) + c$$

Alternative:

$$\text{Let } u = 1 + \sin^2 x \quad du = 2 \sin \cdot \cos x$$

$$\begin{aligned} \int \frac{\sin x \cdot \cos x}{1 + \sin^2 x} &= \frac{1}{2} \int \frac{1}{u} du = \frac{1}{2} \ln|u| + c \\ &= \frac{1}{2} \ln(1 + \sin^2 x) + c \end{aligned}$$



$$12. \int x^5 \cdot \cos(x^3) = \int x^2 \cdot x^3 \cdot \cos(x^3) dx$$

Solution: Let  $u = x^3$   
 $du = 3x^2 dx$

$$\int \underbrace{x^2 x^3 \cos(x^3)}_{u \cdot \cos u} dx = \int u \cdot \cos(u) \frac{du}{3} =$$

$$= \frac{1}{3} \int u \cdot \cos u \, du = \frac{1}{3} \int u \cdot (\sin' u) \, du =$$

$$= \frac{1}{3} \left[ u \cdot \sin(u) - \int u' \sin u \, du \right] = \frac{1}{3} ($$

$$= \frac{1}{3} (u \cdot \sin u - \int \sin u \, du) = \frac{1}{3} (u \cdot \sin u + \cos u) + C = \frac{1}{3} (x^3 \cdot \sin x^3 + \cos(x^3)) + C$$

January 14, 2026

## Trigonometric Integrals

1. Integrals of form  $\int \sin^m x \cos^n x dx$
2. Integrals of form  $\int \tan^m x \sec^n x dx$
3. Integrals of form  $\int \sin(mx) \sin(nx) dx$   
... etc

$$1 \int \sin^m x \cdot \cos^n x dx$$

$$\int \cos^3 x dx = \int (1 - \sin^2 x) \cos x dx =$$

$$\cos^3 x = \cos x \cdot \cos^2 x = \cos x (1 - \sin^2 x)$$

$$\int \underbrace{(1 - \sin^2 x)}_{1-u^2} \underbrace{\cos x dx}_{du} = \int (1 - u^2) du$$

$$\text{let } u = \sin x$$

$$du = \cos x$$

$$\int \sin^m x \cdot \cos^n x \, dx$$

Case 1:  $n$  is odd

$$n = 2k + 1$$

$$\int \sin^m x \cdot \cos^{2k+1} x \, dx = \int \sin^m x \cdot (\cos^2 x)^k \cos x \, dx$$

$$= \int \sin^m x \cdot (1 - \sin^2 x)^k \cos x \, dx$$

Case 2:  $m$  is odd ( $m = 2k + 1$ )

$$\int \sin^{2k+1} x \cdot \cos^n x \, dx = \int \sin^{2k} x \cdot \sin x \cdot \cos^n x \, dx$$

$$= \int (1 - \cos^2 x)^k \cdot \cos^n x \cdot \sin x \, dx \quad u = \cos x$$

Case 3:  $m$  &  $n$  odd, do any

Ex.

$$\int \sin^6 x \cdot \cos^5 x \, dx = \int \sin^6 x \cdot \cos^4 x \cdot \cos x \, dx$$

$$= \int \sin^6 x \cdot (\cos^2 x)^2 \cos x \, dx \quad \begin{array}{l} \text{let } u = \sin x \\ du = \cos x \, dx \end{array}$$

$$= \int \underbrace{\sin^6 x \cdot (1 - \sin^2 x)^2}_{u^6 \cdot (1 - u^2)^2} \underbrace{\cos x \, dx}_{du}$$

$$= \int u^6 (1 - u^2)^2 \, du = \int u^6 (1 - 2u^2 + u^4) \, du$$

$$= \int (u^6 - 2u^8 + u^{10}) \, du = \frac{u^7}{7} - 2 \frac{u^9}{9} + \frac{u^{11}}{11} + C$$

$$= \frac{\sin^7 x}{7} - \frac{2 \sin^9 x}{9} + \frac{\sin^{11} x}{11} + C$$

Ex.  $\int \sin^4 x \cdot \cos^6 x \, dx = \int (\sin^2 x)^2 (\cos^2 x)^3 \, dx$

$$= \int \left( \frac{1 - \cos(2x)}{2} \right)^2 \left( \frac{1 + \cos(2x)}{2} \right)^3 \, dx$$

Ex.  $\int \cos^2 x \, dx$

Solution:

$$\int \cos^2 x \, dx = \int \frac{1 + \cos(2x)}{2} \, dx =$$

$$\frac{1}{2} \left( x + \int \cos(2x) \, dx \right) = \frac{x}{2} + \frac{1}{2} \cdot \frac{\sin(2x)}{2} + C$$

$$\int \cos(2x) = \frac{\sin(2x)}{2}$$

Ex.  $\int \cos^2 x \cdot \sin^2 x \, dx = \int \frac{1 - \cos(2x)}{2} \cdot \frac{1 + \cos(2x)}{2} \, dx$

$$= \frac{1}{4} \int (1 - \cos(2x))(1 + \cos(2x)) \, dx =$$

$$= \frac{1}{4} \int (1 - \cos^2(2x)) \, dx = \frac{1}{4} x - \frac{1}{4} \int \cos^2(2x) \, dx$$

$$= \frac{x}{4} - \frac{1}{4} \int \frac{1 + \cos(4x)}{2} \, dx$$

$$= \frac{x}{4} - \frac{x}{4} - \frac{1}{8} \int \cos(4x) dx$$

$$= \frac{x}{8} - \frac{1}{8} \cdot \frac{\sin(4x)}{4} + C = \boxed{\frac{x}{8} - \frac{\sin(4x)}{32} + C}$$

$$\int \tan^{\overset{\text{odd}}{\textcircled{m}}} x \cdot \sec^{\overset{\text{even}}{\textcircled{n}}} x dx$$

1. If  $n$  is even,  $n = 2k$

$$\int \tan^m x \cdot \sec^{2k} x dx = \int \tan^m x \cdot \underbrace{\sec^{2k-2} x}_{(\sec^2 x)^{k-1}} \cdot \sec^2 x$$

$$= \int \underbrace{\tan^m x (1 + \tan^2 x)^{k-1}}_{\text{terms of } u} \underbrace{\sec^2 x}_{du} dx$$

2. If  $m = 2k+1$

$$\int \tan^{10} x \cdot \sec^6 x dx = \int \tan^{10} x \cdot \sec^4 x \cdot \sec^2 x$$

$$= \int \tan^{10} x (\sec^2 x)^2 \cdot \sec^2 x dx$$

$$= \int \tan^{10} x (1 + \tan^2 x)^2 \cdot \sec^2 x \, dx$$

$$\text{let } u = \tan x$$

$$du = \sec^2 x \, dx$$

$$= \int u^{10} (1 + u^2)^2 \, dx = \int u^{10} (1 + 2u^2 + u^4) \, du$$

$$= \int (u^{10} + 2u^{12} + u^{14}) \, du = \frac{u^{11}}{11} + 2 \frac{u^{13}}{13} + \frac{u^{15}}{15} + C$$

$$= \frac{\tan^{11} x}{11} + \frac{2 \tan^{13} x}{13} + \frac{\tan^{15} x}{15} + C$$

$$\int \cot^3 x \cdot \csc^3 x \, dx$$

Solution:

$$\int \cot^2 x \cdot \csc^2 x (\cot x \cdot \csc x) \, dx =$$

$$\int (\csc^2 x - 1) \csc^2 x \cot x \cdot \csc x \, dx$$

$$\text{let } u = \csc x$$

$$du = -\cot x \csc x \, dx$$

$$= -\int (u^2 - 1) u^2 \, du = -\int (u^4 - u^2) \, du =$$

$$= \int (u^2 - u^4) \, du = \frac{u^3}{3} - \frac{u^5}{5} + C$$

$$= \frac{\csc^3 x}{3} - \frac{\csc^5 x}{5} + C$$

$$1 \int \cos(7x) \cdot \cos(4x) \, dx =$$

$$= \int \frac{1}{2} (\cos(7-4x) + \cos(7x+4x)) \, dx$$

$$= \frac{1}{2} \int (\cos(3x) + \cos(11x)) \, dx$$



$$= \frac{1}{2} \left( \frac{\sin(3x)}{3} + \frac{\sin(11x)}{11} \right) + C$$

# Tutorial

January 16, 2026

$$1. \int x^2 \cdot \tan(x^3 + 1) dx$$

Solution Let  $u = x^3 + 1$

$$du = 3x^2 dx \rightarrow \frac{du}{3} = x^2 dx$$

$$\int x^2 \cdot \tan(x^3 + 1) dx$$

$$= \int \frac{du}{3} \cdot \tan(u) = \frac{1}{3} \int \tan(u) du =$$

$$= \frac{1}{3} \ln|\sec(u)| + C = \frac{1}{3} \ln|\sec(x^3 + 1)| + C$$

$$2. \int \frac{x^9}{4+x^{10}} dx$$

$$\text{Let } u = 4+x^{10}$$

$$du = 10x^9$$

$$x^9 = \frac{du}{10}$$

$$\int \frac{x^9}{4+x^{10}} = \frac{1}{10} \int \frac{1}{u} \cdot du = \frac{1}{10} \ln|u| + C$$

$$= \frac{1}{10} \ln|4+x^{10}| + C$$

$$3. \int \frac{x^4}{4+x^{10}} dx = \int \frac{x^4}{4+(x^5)^2} dx$$

$$\text{Let } u = x^5$$

$$du = 5x^4 dx$$

$$x^4 dx = \frac{du}{5}$$

$$= \frac{1}{5} \int \frac{1}{4+u^2} du = \frac{1}{5} \int \frac{1}{2^2+u^2} du =$$

$$= \frac{1}{5} \cdot \frac{1}{2} \tan^{-1} \left( \frac{u}{2} \right) + C =$$

$$= \frac{1}{5} \cdot \frac{1}{2} \cdot \tan^{-1} \left( \frac{x^5}{2} \right) + C$$

$$4. \int \frac{1}{\sqrt{21+4x-x^2}}$$

Solution

$$21+4x-x^2 = 21-(x^2-4x) =$$

$$= \underline{21} - \underbrace{(x^2 - 2 \cdot x \cdot 2 + 2^2)}_{(x-2)^2} - \underline{2^2} = 25 - (x-2)^2$$

$$\int \frac{1}{\sqrt{25 - (x-2)^2}} dx$$

$$\begin{aligned} \text{Let } u &= x-2 \\ du &= dx \end{aligned}$$

$$= \int \frac{1}{\sqrt{25 - u^2}} du$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} = \sin^{-1}\left(\frac{x}{a}\right) + C$$

$$= \int \frac{1}{\sqrt{5^2 - u^2}} du = \sin^{-1}\left(\frac{u}{5}\right) + C$$

$$= \boxed{\sin^{-1}\left(\frac{x-2}{5}\right) + C}$$

$$6. \int x^5 \sqrt{4 + x^2} dx$$

Solution

$$\text{Let } u = 4 + x^2$$

$$du = 2x dx$$

$$\boxed{x dx = du/2}$$

$$\int x^5 \underbrace{\sqrt{4 + x^2}}_{u^2} dx = \int \underbrace{x \cdot (x^2)^2}_{\text{red circle}} \underbrace{\sqrt{4 + x^2}}_{\text{red circle}} dx \quad \frac{du}{2}$$

$$= \frac{1}{2} \int (u-4)^2 \sqrt{u} \, du \rightarrow \text{brute force}$$

$$= \frac{1}{2} \int (u^2 - 8u + 16) u^{\frac{1}{2}} \, du =$$

$$= \frac{1}{2} \int \left( u^{\frac{5}{2}} - 8u^{\frac{3}{2}} + 16u^{\frac{1}{2}} \right) \, du =$$

$$= \frac{1}{2} \left( \frac{u^{\frac{7}{2}}}{\frac{7}{2}} - 8 \frac{u^{\frac{5}{2}}}{\frac{5}{2}} + \frac{16u^{\frac{3}{2}}}{\frac{3}{2}} \right) + C =$$

$$= \frac{(x^2+4)^{\frac{7}{2}}}{7} - 8 \frac{(x^2+4)^{\frac{5}{2}}}{5} + 16 \frac{(x^2+4)^{\frac{3}{2}}}{3} + C$$

$$7. \int \frac{1}{1+e^{-x}} \, dx = \int \frac{1}{1+\frac{1}{e^x}} \, dx =$$

$$= \int \frac{e^x}{e^x+1} \, dx =$$

$$\text{let } u = e^x + 1$$

$$du = e^x \, dx$$

$$= \int \frac{1}{u} du = \ln|u| + C = \ln(e^x + 1) + C$$

## TRIG INTEGRALS

$$9. \int \tan^{\textcircled{3}} x \cdot \sec^1 x \, dx =$$

odd ✓

$$= \int \tan^2 x \cdot \tan x \cdot \sec x \, dx =$$

$$= \int (\sec^2 - 1) \tan x \cdot \sec x \, dx$$

$$\text{Let } u = \sec x$$

$$du = \tan x \cdot \sec x \, dx$$

$$= \int (u^2 - 1) du = \frac{u^3}{3} - u + C =$$

$$\frac{\sec^3}{3} - \sec x + C$$

$$\begin{aligned}
& \int \cos^4 x \, dx = \int (\cos^2 x)^2 \, dx \\
&= \int \left( \frac{1 + \cos(2x)}{2} \right)^2 \, dx \\
&= \frac{1}{4} \int (1 + \cos(2x))^2 \, dx = \frac{1}{4} \int 1 + 2 \cdot \cos(2x) + \cos^2(2x) \, dx \\
&= \frac{x}{4} + \frac{1}{2} \int \cos(2x) \, dx + \frac{1}{4} \int \cos^2(2x) \, dx \\
&= \frac{x}{4} + \frac{1}{2} \frac{\sin(2x)}{2} + \frac{1}{4} \int \frac{1 + \cos(4x)}{2} \, dx \\
&= \frac{x}{4} + \frac{\sin(2x)}{4} + \frac{1}{8} \int 1 + \cos(4x) \, dx = \\
&= \frac{x}{4} + \frac{\sin(2x)}{4} + \frac{x}{8} + \frac{1}{8} \frac{\sin(4x)}{4} + C \\
&= \frac{3x}{8} + \frac{\sin(2x)}{4} + \frac{\sin(4x)}{32} + C
\end{aligned}$$

January 19, 2026

## Trigonometric Substitution

$$\int x^2 \sqrt{16 - x^2} dx = \int 16 \sin^2 \theta \sqrt{16 - 16 \sin^2 \theta} \cdot 4 \cos \theta d\theta$$

$\Downarrow$   
 $\sqrt{4^2 - x^2}$

$$= 4 \sqrt{1 - \sin^2 \theta} = 4 \sqrt{\cos^2 \theta}$$

$$x = 4 \sin \theta$$

$$= 4 \cos \theta$$

$$dx = 4 \cos \theta d\theta$$

$$= 256 \int \sin^2 \theta \cos^2 \theta d\theta$$

trig integral

1. For  $\sqrt{a^2 - x^2}$

$x = a \cdot \sin \theta$ , choose  $\theta$  such that  
this happens

Is there even such  $\theta$ ?



$$a^2 - x^2 \geq 0 \quad x^2 \leq a^2 \quad -a \leq x \leq a$$

yes, because  $-a \leq x \leq a$

$$\begin{aligned} \sqrt{a^2 - x^2} &= \sqrt{a^2 - a^2 \sin^2 \theta} = a \sqrt{1 - \sin^2 \theta} \\ &= a \sqrt{\cos^2 \theta} = a |\cos \theta| = a \cdot \cos \theta \end{aligned}$$

2.

For  $\sqrt{a^2 + x^2}$  no restrictions

$$\text{let } x = a \cdot \tan \theta$$

$$\begin{aligned} \sqrt{a^2 + x^2} &= \sqrt{a^2 + a^2 \tan^2 \theta} = a \cdot \sqrt{\sec^2 \theta} \\ &= a |\sec \theta| = a \cdot \sec \theta \end{aligned}$$

3.

$$\sqrt{x^2 - a^2}$$

$$x^2 - a^2 \geq 0$$

$$x^2 \geq a^2$$

$$\underline{x \geq a}$$

or

$$\underline{x \leq -a}$$

Ex 1.

$$\int x \sqrt{16-x^2} dx = \int x \sqrt{4^2-x^2} dx$$

$$\text{let } x = 4 \sin \theta \left( -\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2} \right)$$

$$dx = 4 \cos \theta d\theta$$

$$\text{Then } \sqrt{16-x^2} = \sqrt{16-16 \sin^2 \theta} =$$

$$= \sqrt{16(1-\sin^2 \theta)} = 4 \sqrt{\cos^2 \theta} = 4 |\cos \theta|$$

$$= 4 \cos \theta$$

$$\text{because } -\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2}$$

$$\int x \sqrt{16-x^2} dx = 4 \int 4 \sin \theta \cdot 4 \cos \theta \cdot 4 \cos \theta$$

$$= 64 \int \sin \theta \cos^2 \theta d\theta$$

$$\text{let } u = \cos \theta$$

$$du = -\sin \theta d\theta$$

$$= 64 \int \cos^2 \theta \underbrace{\sin \theta}_{-du} d\theta$$

$$= -64 \int u^2 du = \boxed{-\frac{64}{3} u^3 + C}$$

$$= -\frac{64}{3} \cdot \cos^3 \theta + C$$

$$\sin \theta = \frac{x}{4}$$

$$\theta = \sin^{-1}\left(\frac{x}{4}\right)$$

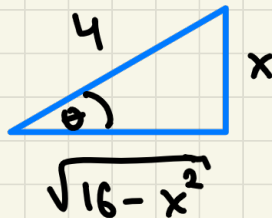
because  $-\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2}$

$$= \boxed{-\frac{64}{3} \cos^3\left(\sin^{-1}\left(\frac{x}{4}\right)\right) + C}$$

If  $x = 4 \sin \theta$ ,  $-\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2}$

compute  $\cos \theta$  in terms of  $x$

$$\sin \theta = \frac{x}{4}$$



$$\cos \theta = \frac{\sqrt{16 - x^2}}{4}$$

Conclusion

$$\int x \sqrt{16-x^2} dx = -\frac{64}{3} \cos^3 \theta + C$$

$$= -\frac{64}{3} \left( \frac{\sqrt{16-x^2}}{4} \right)^3 + C$$

$$= -\frac{(\sqrt{16-x^2})^3}{3} + C$$

2.

$$\int \frac{x^3}{\sqrt{x^2+9}} dx$$

$x^2 + 3^2$

Solution

let  $x = 3 \tan \theta$

$dx = 3 \sec^2 \theta d\theta$

Then  $\sqrt{x^2+9} = \sqrt{9 \tan^2 \theta + 9} = 3 \sqrt{\tan^2 \theta + 1}$

$$= 3 \sqrt{\sec^2 \theta} = 3 |\sec \theta| = 3 \sec \theta$$

$$\int \frac{x^3}{\sqrt{x^2+9}} = \int \frac{27 \tan^3 \theta}{3 \sec \theta} \cdot 3 \sec^2 \theta d\theta$$

$$= 27 \int \tan^3 \theta \sec \theta d\theta$$

$$= 27 \int \tan^3 \theta \cdot \sec \theta \, d\theta$$

$$= 27 \int \tan^2 \theta \cdot \tan \theta \cdot \sec \theta \, d\theta$$

$$= 27 \int (\sec^2 - 1) \tan \theta \cdot \sec \theta \, d\theta$$

$$\text{let } u = \sec \theta$$

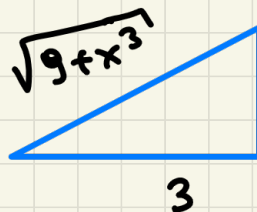
$$du = \tan \theta \sec \theta \, d\theta$$

$$= 27 \int (u^2 - 1) \, du = 27 \left( \frac{u^3}{3} - u \right) + C$$

$$= 9u^3 - 27u + C$$

$$= 9 \sec^3 \theta - 27 \sec \theta + C$$

$$x = 3 \tan \theta$$



$$\tan \theta = \frac{x}{3}$$

$$\sec \theta = \frac{\sqrt{9+x^2}}{3}$$

$$\int \frac{x^3}{\sqrt{x^2+9}} dx = 9 \sec^3 \theta - 27 \sec \theta + C$$

$$= 9 \left( \frac{\sqrt{9+x^2}}{3} \right)^3 - 27 \frac{\sqrt{9+x^2}}{3} + C$$

$$= \left( \frac{\sqrt{9+x^2}}{3} \right)^3 - 9 \sqrt{9+x^2} + C$$

$$3. \int \frac{\sqrt{x^2-1}}{x^6} dx$$

Solution

$$x = \sec \theta$$

$$dx = \tan \theta \sec \theta d\theta$$

$$\sqrt{x^2-1} = \sqrt{\sec^2 \theta - 1} = \sqrt{\tan^2 \theta} = \tan \theta$$

$$\text{Then } \int \frac{\sqrt{x^2-1}}{x^6} dx = \int \frac{\tan \theta}{\sec^6 \theta} \tan \theta \sec \theta d\theta$$

$$= \int \frac{\tan^2 \theta}{\sec^5 \theta} d\theta = \int \frac{\frac{\sin^2 \theta}{\cos^2 \theta}}{\frac{1}{\cos^5 \theta}} d\theta =$$

$$= \int \cos^3 \theta \cdot \sin^2 \theta d\theta = \int \cos^2 \theta \cdot \sin^2 \theta \cos \theta d\theta$$

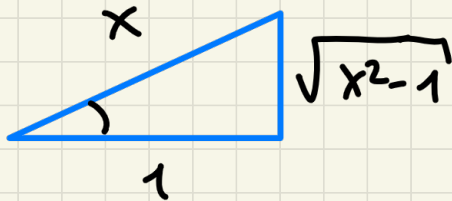
$$= \int (1 - \sin^2 \theta) \sin^2 \theta \cdot \cos \theta d\theta =$$

$$= \int (1 - u^2) u^2 du = \int (u^2 - u^4) du =$$

$$\frac{u^3}{3} - \frac{u^5}{5} + C = \boxed{\frac{\sin^3 \theta}{3} - \frac{\sin^5 \theta}{5} + C}$$

$$\sec \theta = \frac{x}{1}$$

$$\sin \theta = \frac{\sqrt{x^2 - 1}}{x}$$



Then:

$$\int \frac{\sqrt{x^2-1}}{x^6} dx = \frac{\sin^3 \theta}{3} - \frac{\sin^5 \theta}{5} + C$$

$$= \frac{1}{3} \left( \frac{\sqrt{x^2+1}}{x} \right)^3 - \frac{1}{5} \left( \frac{\sqrt{x^2-1}}{x} \right)^5 + C$$